

Project Report - Mr.HelpMate AI (Group Member Life Insurance Policy)

This project presents a Retrieval-Augmented Generation (RAG) based generative search system for the insurance domain.

Using a long life insurance policy document, the system integrates embedding, search, and generation layers to retrieve and generate contextually accurate answers.

Multiple embedding strategies, chunking methods, and model experiments were conducted to enhance retrieval precision and generation quality.

Project Goals

The objective of this project is to design and implement a robust generative search system capable of answering complex questions from a life insurance policy document. The architecture follows the Retrieval-Augmented Generation (RAG) paradigm, incorporating three major layers: Embedding, Search, and Generation. Each layer was tested with multiple strategies to identify the most effective design choices.

Data Source

A single long-form life insurance policy PDF document was used as the primary data source. The document was pre-processed, cleaned, and segmented into text chunks suitable for vector embedding. Text extraction was performed using standard PDF processing libraries, ensuring clean and structured text.

System Architecture Overview

The RAG-based system was structured into three layers:

- **Embedding Layer:** Responsible for text preprocessing, chunking, and embedding generation.
- **Search Layer:** Implements vector-based retrieval using ChromaDB and caching

mechanisms.

- **Generation Layer:** Employs an LLM to synthesize final answers based on retrieved context.

Embedding Layer

The embedding layer plays a crucial role in determining retrieval quality. The document was chunked using overlapping sliding windows of varying sizes (ranging from 300–600 tokens). Several embedding models were tested and compared, including OpenAI's `text-embedding-3-large` and multiple SentenceTransformers models such as `all-MiniLM-L6-v2`, `multi-qa-MPNet-base-dot-v1`, and `paraphrase-MiniLM-L12-v2`. Empirical tests demonstrated that the MPNet model offered a balance between performance and computational efficiency.

Chunking strategies and embedding dimensionality significantly impacted retrieval quality. Experiments revealed that larger chunks captured richer semantic context but reduced retrieval precision, whereas smaller chunks improved relevance at the cost of context completeness.

Search Layer

In the search layer, queries were designed to evaluate the system's ability to locate specific policy details. Three representative queries were selected:

1. What are conditions to form Civil Union?
2. In what conditions the policy can be terminated?
3. What is the claim process in terms of accidental death?

The system employed ChromaDB as the vector store. Each query was embedded using the same model as the chunks and matched against the database using cosine similarity. A caching mechanism was implemented to store embeddings and search results, reducing redundant computation. To further refine the retrieval, a re-ranking block using a cross-

encoder model (`cross-encoder/ms-marco-MiniLM-L-6-v2`) was incorporated to improve top-3 search relevance.

t8886/notebooks/OneDrive/Shreya_C71/HelpMate_AI/HelpMate_AI_Assignment.ipynb

Jupyter HelpMate_AI Assignment Last Checkpoint: 10 minutes ago

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Python 3 (ipykernel) Not Trusted

Semantic Search Pipeline

```
[90]: # Read the user query
query = input()

What are condition to form civil union

[91]: cache_results = cache_collection.query(
    query_text=query,
    n_results=1
)

[92]: cache_results

[92]: {'ids': [[]],
      'embeddings': None,
      'documents': [[]],
      'uris': None,
      'included': ['metadatas', 'documents', 'distances'],
      'metadata': None,
      'metadatas': [[]],
      'distances': [[]]}

[93]: results = insurance_collection.query(
    query_text=query,
    n_results=10
)

[94]: results

[94]: {'ids': [['7', '29', '8', '6', '23', '24', '30', '14', '34', '38']],
      'embeddings': None,
      'documents': [['The legally recognized union of two eligible individuals of the same sex established according to law. Civil Union Partner for two persons to establish a Civil Union in Rhode Island, it shall be necessary that they satisfy all of the following criteria: a. not be a party to another Civil Union or marriage in Rhode Island; b. be of the same sex and therefore be excluded from the marriage laws of Rhode Island or any other state; c. be at least 18 years of age; d. not be related to the other proposed party to the Civil Union. NOTE: For the purposes of this Group Policy, the term "spouse" will include Civil Union Partner, except as otherwise provided in this Group Policy. Date of Issue The date this Group Policy is placed in force: November 1, 2007. Dependent a. A Member's spouse, if that spouse: (1) is legally married to the Member; and (2) is not in the Armed Forces of any country; and (3) is not insured under this Group Policy as a Member. A Member's spouse will also include a Civil Union Partner. b. A Member's Dependent Child (or Children) as defined below. Dependent Child; Dependent Children a. A Member's natural child or stepchild, if that child is 0 days but less than 26 years of age. b. A Member's foster child, if that child: (1) meets the requirements above; and This policy has been updated effective January 1, 2014 GC 6002 PART I - DEFINITIONS, PAGE 2',
                    '(1) marriage or establishment of a Civil Union partnership or divorce or termination of a Civil Union partnership; (2) death of a spouse or child; (3) birth or adoption of a child; (4) termination of employment by the Member's spouse or a change in the spouse's employment that causes loss of group coverage; (5) the Member's employment or the Member's spouse's employment changes from part-time to full-time or from full-time to part-time; (6) the Member or the Member's spouse takes an unpaid leave of absence. A change in the Scheduled Benefits because of a request by the Member when a change in...']]
```

Not found in cache. Found in main collection.

109]:

	Metadata	Document	Distance	ID
0	{'Preview_Text': 'T he Principal may terminate...	T he Principal may terminate the Policyholder'...	0.719092	21
1	{'Preview_Text': 'Section C - Policy Terminati...	Section C - Policy Termination Article 1 - Fai...	0.737443	20
2	{'Page_No': 'Page 35', 'Preview_Text': 'Sectio...	Section C - Individual Terminations Article 1 ...	0.844797	32
3	{'Text_Length': 333, 'Page_No': 'Page 36', 'Pr...	A Member's insurance under this Group Policy f...	0.850975	33
4	{'Text_Length': 392, 'Page_No': 'Page 43', 'Pr...	Any individual policy issued will then be in f...	0.857854	40
5	{'Text_Length': 179, 'Page_No': 'Page 45', 'Pr...	(1) If termination is as described in b. (1) a...	0.904881	42
6	{'Preview_Text': 'b. a business assignment; or...	b. a business assignment; or c. full-time stud...	0.934197	34
7	{'Preview_Text': 'PART II - POLICY ADMINISTRAT...	PART II - POLICY ADMINISTRATION Section A - Co...	0.992520	13
8	{'Page_No': 'Page 6', 'Preview_Text': 'TABLE O...	TABLE OF CONTENTS PART I - DEFINITIONS PART II...	1.016217	3
9	{'Preview_Text': 'I f coverage for a Member or...	I f coverage for a Member or Dependent termina...	1.016423	38

Top 3 Search Results for Query 1 → What are conditions to form Civil Union?

Jupyter HelpMate_AI_Assignment Last Checkpoint: 12 minutes ago

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JupyterLab Python 3 (ipykernel)

```

Semantic Search Pipeline

[95]: # Read the user query
query = input()

In what conditions the policy can be terminated

[96]: cache_results = cache_collection.query(
      query_texts=query,
      n_results=1
      )

[97]: cache_results

[97]: {'ids': [],
      'embeddings': None,
      'documents': [],
      'uris': None,
      'included': ['metadatas', 'documents', 'distances'],
      'data': None,
      'metadatas': [],
      'distances': []}

[98]: results = insurance_collection.query(
      query_texts=query,
      n_results=10
      )

[99]: results

[99]: {'ids': [['21', '20', '32', '33', '40', '42', '34', '13', '3', '38']],
      'embeddings': None,
      'documents': [['T he Principal may terminate the Policyholder's coverage on any premium due date if the Policyholder relocates to a state where this G
roup Policy is not marketed, by giving the Policyholder 31 days advanced notice in Writing. Article 4 - Policyholder Responsibility to Members If this G
roup Policy terminates for any reason, the Policyholder must: a. notify each Member of the effective date of the termination; and b. refund or otherwi
se account to each Member all contributions received or withheld from Members for premiums not actually paid to The Principal. This policy has been upd
ated effective January 1, 2014 PART II - POLICY ADMINISTRATION GC 6005 Section C - Policy Termination, Page 2",
"Section C - Policy Termination Article 1 - Failure to Pay Premium This Group Policy will terminate at the end of the Grace Period if total premium
due has not been received by The Principal before the end of the Grace Period. Failure by the Policyholder to pay the premium within the Grace Period w
ill be deemed notice by the Policyholder to The Principal to discontinue this Group Policy at the end of the Grace Period. Article 2 - Termination Righ
ts of the Policyholder The Policyholder may terminate this Group Policy effective on the day before any premium due date by giving Written notice to Th
e Principal prior to that premium due date. The Policyholder's issuance of a stop-payment order for any amounts used to pay premiums for the Policyhold
er's coverage will be considered Written notice from the Policyholder. Article 3 - Termination Rights of The Principal The Principal may nonrenew or te
rminate this Group Policy by giving the Policyholder 31 days advance notice in Writing, if the Policyholder: a. ceases to be actively engaged in busine
ss for profit within the meaning of the Internal Revenue Code, or be established as a legitimate nonprofit corporation within the meaning of the Intern
al Revenue Code; or b. fails to maintain the participation percentages requirements of PART II, Section A with respect to eligible employees, excluding
those for whom Proof of Good Health is not satisfactory to The Principal; or c. fails to maintain three or more insured employees under this Group Pol

```

Not found in cache. Found in main collection.

109]:

	Metadatas	Documents	Distances	IDs
0	{'Preview_Text': 'T he Principal may terminate...	T he Principal may terminate the Policyholder'...	0.719092	21
1	{'Preview_Text': 'Section C - Policy Terminati...	Section C - Policy Termination Article 1 - Fai...	0.737443	20
2	{'Page_No': 'Page 35', 'Preview_Text': 'Sectio...	Section C - Individual Terminations Article 1 ...	0.844797	32
3	{'Text_Length': 333, 'Page_No': 'Page 36', 'Pr...	A Member's insurance under this Group Policy f...	0.850975	33
4	{'Text_Length': 392, 'Page_No': 'Page 43', 'Pr...	Any individual policy issued will then be in f...	0.857854	40
5	{'Text_Length': 179, 'Page_No': 'Page 45', 'Pr...	(1) If termination is as described in b. (1) a...	0.904881	42
6	{'Preview_Text': 'b. a business assignment; or...	b. a business assignment; or c. full-time stud...	0.934197	34
7	{'Preview_Text': 'PART II - POLICY ADMINISTRAT...	PART II - POLICY ADMINISTRATION Section A - Co...	0.992520	13
8	{'Page_No': 'Page 6', 'Preview_Text': 'TABLE O...	TABLE OF CONTENTS PART I - DEFINITIONS PART II...	1.016217	3
9	{'Preview_Text': 'I f coverage for a Member or...	I f coverage for a Member or Dependent termina...	1.016423	38

Top 3 Search Results for Query 2 → In what conditions the policy can be terminated?

localhost:8888/notebooks/OneDrive/Shreya_C71/HelpMate_AI/HelpMate_AI_Assignment.ipynb

Jupyter HelpMate_AI_Assignment Last Checkpoint: 15 minutes ago

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JupyterLab Python 3 (ipykernel)

In case it doesn't the query then passes to the financedata_collection, and the query results get appended to cache_collection (embeddings, metadata, etc)

Semantic Search Pipeline

```
[100]: # Read the user query
query = input()

What is the claim process in terms of accidental death

[101]: cache_results = cache_collection.query(
        query_texts=query,
        n_results=1
    )

[102]: cache_results

[102]: {'ids': [],
      'embeddings': None,
      'documents': [],
      'uris': None,
      'included': ['metadatas', 'documents', 'distances'],
      'data': None,
      'metadatas': [],
      'distances': []}

[103]: results = insurance_collection.query(
        query_texts=query,
        n_results=10
    )

[104]: results

[104]: {'ids': [['52', '51', '59', '5', '58', '50', '54', '53', '48', '44']],
      'embeddings': None,
      'documents': [['Exposure Exposure to the elements will be presumed to be an injury if: a. such exposure is due to an accidental bodily injury; and b. within 365 days after the injury, the Member incurs a loss that is the result of the exposure; and c. this Group Policy would have covered the injury resulting from the accident. Article 4 - Seat Belt/Airbag Benefit If the Member loses his or her life as a result of an accidental injury sustained while driving or riding in an Automobile, an additional benefit of $10,000 will be paid to the beneficiary named for Member Life Insurance, provided all Benefit Qualifications as described in Article 2 are met and: a. the Automobile is equipped with factory-installed Seat Belts; and b. the Seat Belt was in actual use by the Member and properly fastened at the time of the accident; and c. the position of the Seat Belt is certified in the official report of the accident or by the investigating officer. This additional benefit payment will also apply if the Member was driving an Automobile equipped with a properly functioning driver-side air bag or riding as a passenger in an Automobile equipped with a properly functioning passenger-side air bag, although the Member's Seat Belt may not have been fastened at the time of the accident. The properly functioning and/or deployment of the air bag must be certified in the official report of the accident or by the investigating officer. For the purpose of this benefit, "Automobile" means a four-wheel passenger vehicle, station wagon, pick-up truck, or van-type vehicle, but excludes recreational-type vehicles such as a "dune-buggy" or an "all-terrain" vehicle. The term "Seat Belt" means a factory-installed device that forms an occupant restraint and injury avoidance system. Article 5 - Loss of Use or Partialysis Benefit This policy has been updated effective January 1, 2014 PART IV - BENEFITS GC 6015 Section B - Member Accidental Death and Dismemberment
```

```
results_df = pd.DataFrame({
    'IDs': ids,
    'Documents': documents,
    'Distances': distances,
    'Metadatas': metadatas
})

results_df
```

Not found in cache. Found in main collection.

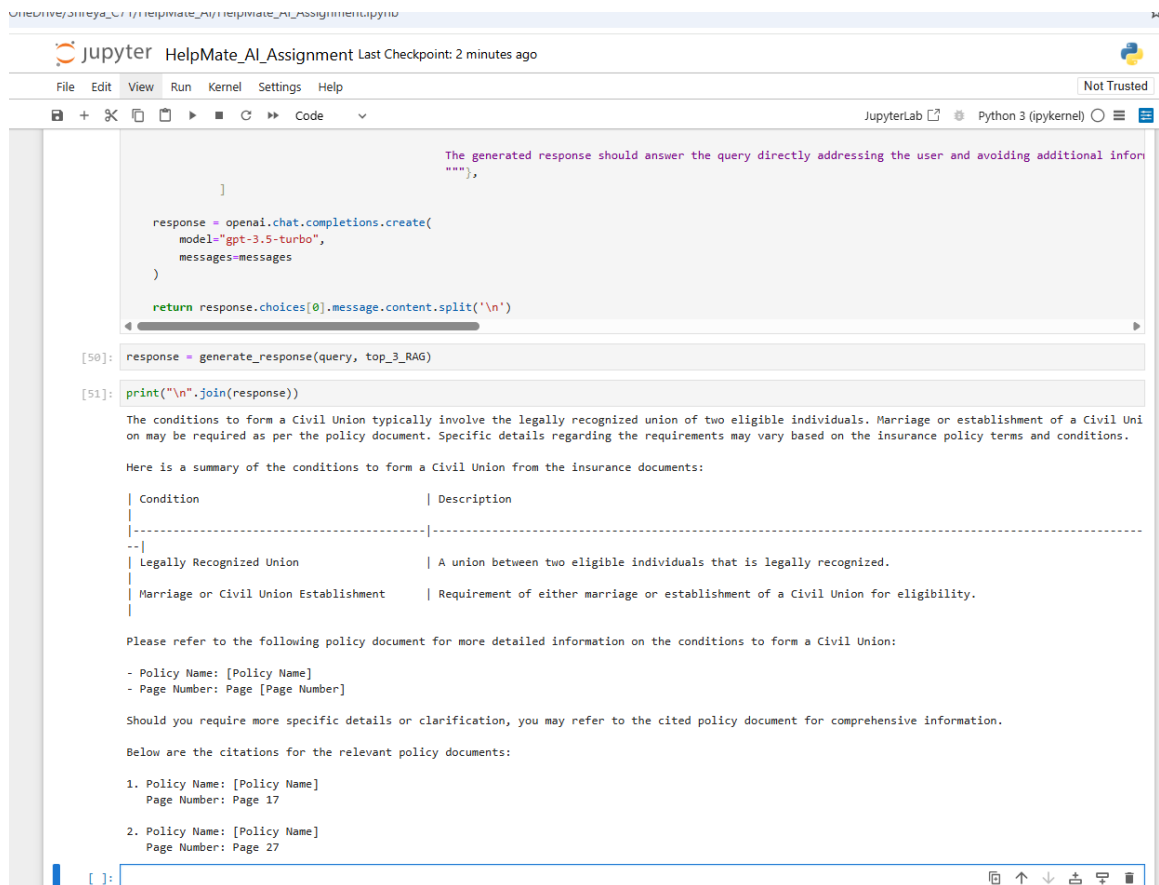
105]:

	Metadatas	Documents	Distances	IDs
0	{'Preview_Text': 'Exposure Exposure to the ele...	Exposure Exposure to the elements will be pres...	0.868915	52
1	{'Preview_Text': 'f. claim requirements liste...	f. claim requirements listed in PART IV, Sect...	0.873304	51
2	{'Text_Length': 322, 'Preview_Text': 'A claima...	A claimant may request an appeal of a claim de...	0.904050	59
3	{'Text_Length': 171, 'Page_No': 'Page 8', 'Pre...	Section A - Member Life Insurance Schedule of ...	0.949321	5
4	{'Preview_Text': 'Section D - Claim Procedures...	Section D - Claim Procedures Article 1 - Notic...	1.003704	58
5	{'Page_No': 'Page 53', 'Preview_Text': 'Sectio...	Section B - Member Accidental Death and Dismem...	1.017759	50
6	{'Text_Length': 321, 'Preview_Text': '% of Sch...	% of Scheduled Covered Loss Benefit Loss of Sp...	1.033374	54
7	{'Page_No': 'Page 56', 'Text_Length': 307, 'Pr...	If a Member sustains an injury, and as a resul...	1.055081	53
8	{'Text_Length': 273, 'Page_No': 'Page 51', 'Pr...	Coverage During Disability will cease on the e...	1.056784	48
9	{'Preview_Text': 'M ember's death, the Death B...	M ember's death, the Death Benefits Payable ma...	1.068380	44

Top 3 Search Results for Query 3

Generation Layer

The generation layer used the retrieved top chunks to construct an augmented prompt for the Large Language Model. Prompt design was critical in guiding the model to produce concise and factual answers. Experiments included few-shot prompting and variations in context length. OpenAI's GPT-based models were tested, with the final version achieving the best factual accuracy and fluency using GPT-4.



```
from openai import OpenAI
client = OpenAI(api_key=api_key)

def generate_response(query, top_3_RAG):
    """Generate a response to the query using the retrieved top 3 chunks from the RAG system.

    The generated response should answer the query directly addressing the user and avoiding additional information.

    """
    messages = [
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": query}
    ]
    response = openai.chat.completions.create(
        model="gpt-3.5-turbo",
        messages=messages
    )
    return response.choices[0].message.content.split('\n')

[50]: response = generate_response(query, top_3_RAG)

[51]: print("\n".join(response))

The conditions to form a Civil Union typically involve the legally recognized union of two eligible individuals. Marriage or establishment of a Civil Union may be required as per the policy document. Specific details regarding the requirements may vary based on the insurance policy terms and conditions.

Here is a summary of the conditions to form a Civil Union from the insurance documents:

| Condition | Description |
|-----|-----|
| Legally Recognized Union | A union between two eligible individuals that is legally recognized. |
| Marriage or Civil Union Establishment | Requirement of either marriage or establishment of a Civil Union for eligibility. |

Please refer to the following policy document for more detailed information on the conditions to form a Civil Union:

- Policy Name: [Policy Name]
- Page Number: Page [Page Number]

Should you require more specific details or clarification, you may refer to the cited policy document for comprehensive information.

Below are the citations for the relevant policy documents:

1. Policy Name: [Policy Name]
   Page Number: Page 17

2. Policy Name: [Policy Name]
   Page Number: Page 27
```

Generated Answer for Query 1

```

    """
    The generated response should answer the query directly addressing the user and avoiding additional info
    """
}

response = openai.chat.completions.create(
    model="gpt-3.5-turbo",
    messages=messages
)

return response.choices[0].message.content.split('\n')

```

[70]: `response = generate_response(query, top_3_RAG)`

[71]: `print("\n".join(response))`

Upon termination, the policy can be ended in the following conditions as per the documents available:

1. Failure to pay the premium by the Principal can lead to termination.
2. Individual terminations are possible under certain circumstances mentioned in the policy document.

Unfortunately, further details on specific conditions for policy termination are not available in the extracted text. To find more specific information, kindly refer to the mentioned policy documents and search for "Policy Termination" and "Individual Terminations" sections for detailed conditions.

Citations:

1. Policy Name: Section C - Policy Termination
Page Number: Page 35
2. Policy Name: Section C - Individual Terminations
Page Number: Page 43

Generated Answer for Query 2

```
5]: response = generate_response(query, top_3_RAG)

5]: print("\n".join(response))

The claim process for accidental death is outlined in Section B of the policy document. It specifies the requirements and steps for initiating a claim in case of accidental death. The document provides detailed information on how to proceed with the claim submission and the essential documents needed for processing the claim. For specific details on the claim process, please refer to the policy document Section B - Member Accidental Death and Dismemberment.

Citation:
- Policy Name: Section B - Member Accidental Death and Dismemberment
- Page Number: Page 53

]:
```

Generated Answer for Query 3

Results

Across several experiments, we evaluated how different embeddings and chunk sizes affected the system’s performance.

OpenAI embeddings provided the most accurate semantic matches but required more computational resources. In contrast, SentenceTransformers offered a practical balance between accuracy and efficiency, making them suitable for offline or resource-limited setups.

Additionally, applying a re-ranking step further improved the precision of the retrieved results.

Challenges & Learnings

Key challenges included cleaning unstructured PDF text, tuning chunk sizes, and balancing retrieval precision with recall. Designing a robust prompt for consistent LLM responses required iterative refinement. Additionally, ensuring reproducibility and caching efficiency was critical in optimizing the system's performance.

Conclusion

This project demonstrates the potential of Retrieval-Augmented Generation for domain-specific document search.

Semantic search retrieves the most relevant text passages based on meaning, but its answers are limited to the retrieved content itself.

In contrast, Retrieval-Augmented Generation (RAG) goes a step further—it not only finds relevant information but also uses a language model to generate a coherent, context-aware response.

This makes RAG more effective for answering complex or multi-step queries, while pure semantic search excels in quick, factual retrieval.

References

1. OpenAI Embeddings API Documentation
2. SentenceTransformers Library, HuggingFace
3. ChromaDB Vector Database
4. Cross-Encoder Models, HuggingFace Transformers